

3.3 Solution Architecture

The schematic below illustrates the complete end-to-end Solution Architecture for SCMS, showing how presentation layers, application configurations, platform automation services, and data models integrate on the ServiceNow core.

Architectural Breakdown

1. **Presentation & User Experience Layer:**
 - **Service Portal Interface:** Provides self-service catalog forms for students to submit leave requests and lodge campus complaints.
 - **Agent Workspace / Standard UI:** Gives HODs and Support Techs dedicated views for managing approval tasks and resolving maintenance tickets.
2. **Application Configuration & Business Logic Layer:**
 - **Client-Side Scripts & UI Policies:** Execute real-time form validation (e.g., verifying `End Date >= Start Date`) directly in the browser.
 - **Server-Side Business Rules & Script Includes:** Enforce database integrity, prevent duplicate course registrations, and recalculate student leave balances.
3. **Platform Core & Automation Layer:**
 - **Flow Designer Engine:** Automates approval routing, status propagation, task creation, and email notifications.
 - **ACL Security Matrix:** Enforces role-based read/write access across all `x_scms_*` tables.
4. **Data Storage & Data Model Layer:**
 - Custom transactional and master data tables (`x_scms_student`, `x_scms_faculty`, `x_scms_course`, `x_scms_enrollment`, `x_scms_leave_request`, `x_scms_complaint`).